

Measurement Validity and Scorer Validation

Session 4

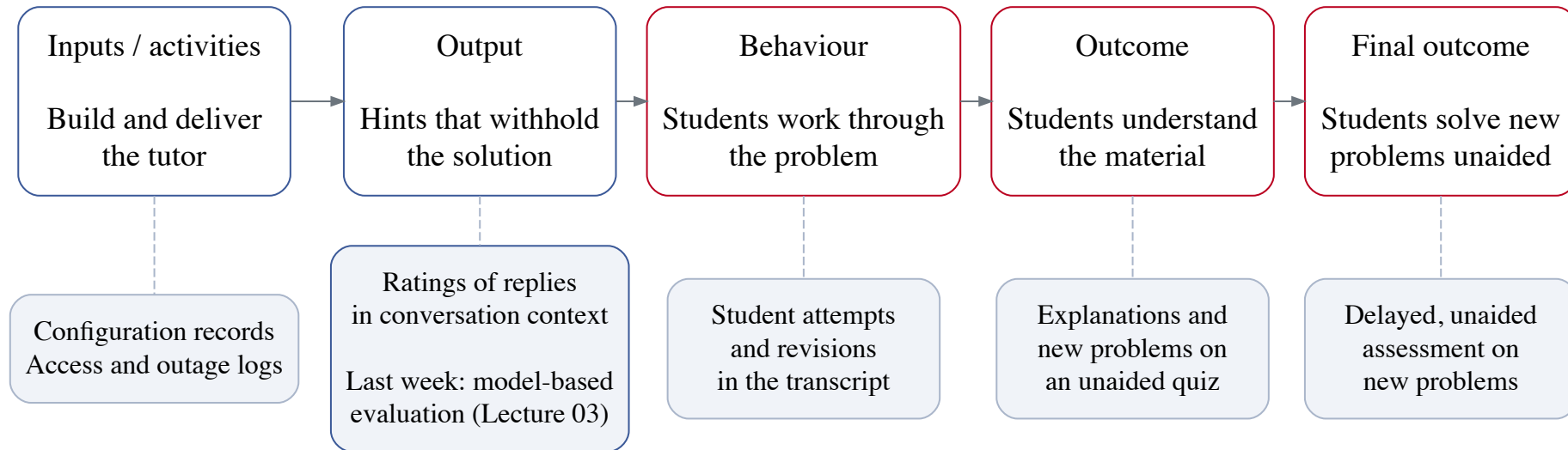
2026-09-22

Developing a measurement plan



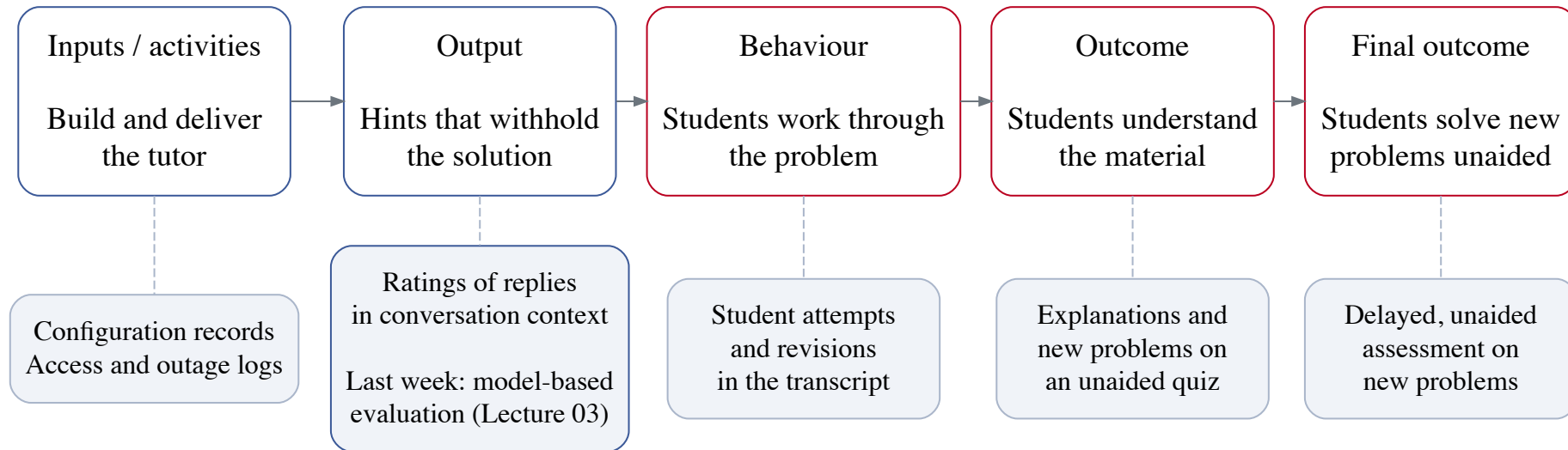
The tutor's Theory of Change

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Student attempts show **effort** (behavior); an unaided quiz tests **understanding** (final outcome).



Learning gains can come with costs

Even if the tutor produces the intended learning, we could find:

Possible tradeoff	Additional measures
Students learn more, but spend much longer on the assignment	Time spent on the assignment, including work outside the tutor
Working through difficulties builds understanding, but increases frustration or reduces confidence	Reported frustration and confidence before and after the assignment
Well-prepared students benefit, while students needing more support fall behind	Prior preparation; learning gains and task completion by preparation level



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⇒ The measurement plan should capture ways that things can go wrong.



Measurements make a theory of change testable

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Different observations point to different explanations:

- › The tutor supplied solutions: the intended help was not delivered.
- › Students rarely opened it: the proposed learning process had little exposure.
- › Students used hints and attempted the work: investigate whether that practice produced understanding.



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The chain helps **locate a possible failure**.

- › Students who use more hints may already be more motivated.
 - » Attributing learning to hint use requires a causal design.



Defining what the score should mean



A construct needs a definition

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The ability to explain a method, choose when to use it, and apply it to a new problem without assistance.



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A rubric breaks the definition down into its component pieces each to be measured individually.



From construct to indicator to score

Step	In the unaided assessment
Construct: the ability we want to infer	Independent understanding
Indicators: observable evidence of that ability	An appropriate estimator choice, a justification tied to the scenario, and a correct application
Scores: rules applied to that evidence	Rate choice, justification, and application separately: 0 = absent or incorrect, 1 = partly correct, 2 = correct



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- › The **intervention instrument** (Session 2) specifies how help is delivered: model, prompt, and access rules.
- › The **measurement instrument** specifies how understanding is assessed: tasks, unaided conditions, and scoring rules.



Measurement makes assumptions about the construct

We infer unobservable constructs from observable properties through a **measurement model**.

Jacobs, A. Z., & Wallach, H. (2021). [Measurement and Fairness](#). *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*, 375–385. ACM. §§2–3.



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- › Combining them assumes a relationship between those indicators and the construct.
- › Recording them introduces assumptions about measurement error.

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Mismatches can produce unfair decisions when the same proxy means different things for different groups.



The mismeasure of man: height and socioeconomic status

Height

Target: a person's physical height.

Measure: a ruler reading.

Choices: standing or seated? Shoes included?

Error: ruler alignment or self-reporting can shift the measurement.

$$\hat{h} = h + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

The reading $\hat{h}$ is true height h plus assumed zero-mean noise.



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Socioeconomic status

Target: social and economic position.

Measure: income as a proxy.

Choices: how should wealth, education, and occupation enter?

Error: equal incomes can conceal very different resources.

$$s = i, \quad \hat{i} = i \Rightarrow s = \hat{i}$$

This assumes SES s equals income i , measured without error as $\hat{i}$.



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› Measuring income still leaves the link between income and SES to justify.

› Alternative approaches use *indices* which we will discuss later in the term.

Jacobs, A. Z., & Wallach, H. (2021). Measurement and Fairness. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)*, 375–385. ACM. §§2.1–2.2, 3.2.2.

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Construct underrepresentation

- › The quiz names the estimator and asks students to substitute numbers into a formula.
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⇒ If the tutor teaches students to reproduce polished explanations, scores could rise without better reasoning on new problems.

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Build a case for the measure

Pilot the assessment with students from the intended population.

Evidence	A concrete check
Content coverage	Map tasks to choosing, explaining, and applying methods across new scenarios. Have instructors review the coverage.
Response processes	Ask pilot students to think aloud: does a correct answer come from reasoning or a memorized cue?
Convergent evidence	Compare scores with a separately scored oral assessment of the same skills. Do they broadly agree?
Discriminant evidence	Rescore answers with the same reasoning but different prose fluency. Do scores stay similar?



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Use the findings to revise the tasks and rubric, or narrow the claim the scores support.



Developing a scoring rubric



Define the construct, then choose the evidence

Our construct is **instructional support**: accurate, relevant guidance that helps the student take the next step while leaving them work to do.

Consider assessing this in **one tutor reply in conversation context**.

- › Show the assignment, a reference solution, and the conversation through that reply.
- › Rate correctness and instructional usefulness of the selected reply.
- › Rate answer disclosure cumulatively: what has the tutor supplied or confirmed so far?



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Hide the model name and experimental condition from raters.

Give human raters and automated scorers the same evidence packet.



Separate the dimensions that matter

Dimension	Scoring rule
Disclosure D: how much of the solution is supplied?	0 = none; 1 = hint; 2 = partial solution; 3 = complete solution
Correctness C: are the claims and steps accurate?	0 = major error; 1 = minor error that leaves the approach usable; 2 = no substantive error
Usefulness U: does the reply support the student's next step?	0 = no actionable support; 1 = relevant but poorly targeted support; 2 = an actionable next step matched to the student's difficulty



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“I cannot help with that” receives $D = 0$, $U = 0$, and $C = \text{N/A}$: it contains no assessable mathematical claim.

Withholding the answer alone earns no credit for usefulness.



Anchor each level in observable evidence

Task: solve $2x + 3 = 11$. The student has not yet attempted a step.

Scoring disclosure:

D	Observable anchor	Example tutor reply
0	Supplies no solution step or direction	"What have you tried?"
1	Suggests a move; leaves its execution to the student	"Try subtracting 3 from both sides."
2	Executes part of the solution; leaves a required step	"Subtracting 3 gives $2x = 8$. What comes next?"
3	States the final answer or supplies all required steps	" $x = 4$."



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Assign the highest level supported by the tutor's contributions so far.

These are **ordered categories**; the gaps between levels have no defined size.



Boundary cases expose missing rules

Add explicit decisions to the scoring manual:

Boundary case	Rule for this rubric
Student asks “Is $x = 4$?”; tutor says “Yes”	Confirmation of a final answer counts as $D = 3$. The student’s answer alone does not.
Tutor says “ $x = 8/2$ ”	An equivalent expression for the final value counts as $D = 3$.
Tutor says “ $x = 5$ ”	A purported final answer counts as $D = 3$; the error also lowers correctness.
Tutor spreads the solution across several replies	Combine its contributions through the selected reply; together they can reach $D = 3$.



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This definition measures **how much the tutor completes the task**, including incorrect attempts and answer verification.



Combining scores changes what counts as good tutoring

- › Suppose a reply gives away the answer but also offers accurate, useful guidance:
 - » $D = 3, C = 2, U = 2$.

How we report the scores	Result for this reply
Keep the dimensions separate	Complete disclosure; fully correct; useful. There is no single overall grade.
Average three scores, each out of 100	Answer withholding: 0; correctness: 100; usefulness: 100. Equal weights give 67/100 .
Require minima to be met: $D \leq 1, C = 2, U \geq 1$	Fail: despite other strengths.



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- › An average lets correctness and usefulness **offset answer disclosure**.
 - » A minimum requirement makes limited disclosure **mandatory**.
- › Choose this rule before comparing systems.



Pilot, discuss, revise, then freeze

1. Select a small pilot with ordinary replies and the boundary cases.
2. Have at least two raters score independently and mark the evidence behind each rating.
3. Compare disagreements: missing context, unclear anchors, or different interpretations of the task?
4. Revise the rules and add examples that distinguish adjacent levels.
5. Freeze a version of the rubric and assess agreement on fresh cases.



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Keep the original independent ratings. Consensus after discussion records a resolved judgment; **fresh independent ratings test reproducibility.**



Assessing reliability



Reliability and validity answer different questions

- › **Validity:** what evidence supports interpreting that score as reflecting the construct?
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- › That agreement leaves the link between prose and understanding unjustified.
- › We need **reproducible judgments** about **evidence relevant to the construct**.



Build the reliability sample

- › Sample tutor replies from the tasks, students, and systems the evaluation will cover.
- › Give raters the same evidence packet and frozen rubric. Score independently.
- › Store one row per item and one column per rater, separately for each dimension.
- › Plan overlapping assignments so each reliability item has at least two ratings.
 - » Three is even better (breaks ties!)



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- › Report a deliberately enriched sample of boundary cases separately
 - » The distribution you sample is the distribution over tasks you evaluate!



Start with the disagreements

For this worked example, collapse disclosure to **complete** ($D = 3$) or **incomplete** ($D < 3$).

Item	Rater A	Rater B
1	Incomplete	Incomplete
2	Incomplete	Incomplete
3	Incomplete	Complete
4	Complete	Complete
5	Complete	Complete
6	Complete	Complete

A ↓ / B →	Incomplete	Complete
Incomplete	2	1
Complete	0	3

- › Raw agreement: $5/6 = 83\%$.
- › Complete: A gives $3/6$; B gives $4/6$.
- › Pooled ratings: **5 incomplete, 7 complete.**



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 - › Pooled ratings: **5 incomplete, 7 complete.**
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- › On item 3 (“ $x=8/2$ ”), the raters disagree about an equivalent final answer.
 - » Our rubric classified this as complete disclosure.



Agreement needs a reference point

In a different sample of **100 replies**, each rater marks only five as complete disclosure.

A ↓ / B →	Incomplete	Complete	Total
Incomplete	90	5	95
Complete	5	0	5
Total	95	5	100

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- › **90% agreement**, entirely on incomplete disclosure. The raters identify **different replies** as complete disclosure.
- › Randomly assigning labels with these 95% / 5% rates would already give about **90.5% agreement**.
- › The category frequencies determine how demanding the reference point is.

Hallgren (2012), [Computing Inter-Rater Reliability for Observational Data](#), "Kappa for Nominal Variables."



Krippendorff's alpha compares two disagreements

$$\alpha = 1 - \frac{D_o}{D_e}$$

- › D_o : observed disagreement between ratings of the **same item**.
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- › If every rating is the same category, $D_e = 0$: α is undefined

Krippendorff, [Computing Krippendorff's Alpha-Reliability](#), §§A–C.



Calculate alpha on a small example

- › Return to the **six items**: five agreements and one disagreement.
- › For nominal labels, a mismatch costs 1; a match costs 0.

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- › All **12 ratings** are pairable: each has another rating on the same item.
- › The pool contains **5 incomplete** and **7 complete** ratings:

$$D_e = \frac{5}{12} \frac{7}{11} + \frac{7}{12} \frac{5}{11} = \frac{70}{132} \approx 0.530$$



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$$\alpha = 1 - \frac{1/6}{70/132} \approx 0.686$$

- › Disagreement is about **69% lower than the pooled reference**.

Krippendorff, [Computing Krippendorff's Alpha-Reliability](#), §A. Worked example uses illustrative tutor ratings.



The measurement scale defines disagreement

The same disagreement rule enters both D_o and D_e .

Scale	Disagreement rule	For our scores
Nominal	Every unequal pair costs 1	Complete vs. incomplete disclosure
Ordinal	Squared distance between categories' average ranks across all ratings	Ordered disclosure levels 0, 1, 2, 3
Interval	Squared numerical difference, $(a - b)^2$	A score with defensible equal units



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- › The labels 0, 1, 2, 3 establish an order. They do not establish equal gaps in disclosure.
- › Choose the scale **before inspecting the coefficient**.



Alpha can use overlapping rater assignments

Item	Rater A	Rater B	Rater C
1	0	0	—
2	—	1	2
3	3	3	2
4	1	—	—

- › Items 1–3 contribute within-item comparisons; item 4 has no pair.
- › With m ratings on an item, each ordered pair receives weight $1/(m - 1)$.
- › This gives the item total weight m , rather than $m(m - 1)$.



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- › With m ratings on an item, each ordered pair receives weight $1/(m - 1)$.
- › This gives the item total weight m , rather than $m(m - 1)$.
- › Missing ratings are allowed, but be careful.
 - » If raters skip difficult cases, agreement on the remaining cases may overstate reproducibility.



What the agreement coefficients measure

- › **Cohen's kappa:** $\kappa = (p_o - p_e) / (1 - p_e)$
 - » p_o : observed agreement; $p_e = \sum_c p_{Ac} p_{Bc}$: expected agreement from each rater's category proportions.
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- › **Intraclass correlation (ICC)**
 - » Share of score variance attributable to item differences under an item–rater model.
 - » **Use for:** numeric scores with meaningful distances, from one rater or an average of raters.
 - » **Absolute agreement** penalizes systematic rater offsets; **consistency** allows them.



ICC: absolute agreement or consistency?

$$y_{ij} = \mu + a_i + b_j + e_{ij}$$

- › Score = overall mean + **item effect** a_i + **rater offset** b_j + **residual** e_{ij} .
 - » Suppose ANOVA (via random effects model) estimates variances as **9**, **4**, and **3**, respectively.

Absolute agreement

- › Raters should give the **same scores**.
 - » A rater who always adds 2 lowers agreement.
- › **Use when:** the score level matters, such as a pass threshold.

$$\text{ICC} = \frac{9}{9 + 4 + 3} \approx 0.56$$

Hallgren (2012), [Computing Inter-Rater Reliability for Observational Data](#), “Common ICC variants.” Illustrative variances; two-way random-effects, single-rater ICCs.



ICC: absolute agreement or consistency?

$$y_{ij} = \mu + a_i + b_j + e_{ij}$$

- › Score = overall mean + **item effect** a_i + **rater offset** b_j + **residual** e_{ij} .
 - » Suppose ANOVA (via random effects model) estimates variances as **9**, **4**, and **3**, respectively.

Absolute agreement

- › Raters should give the **same scores**.
 - » A rater who always adds 2 lowers agreement.
- › **Use when:** the score level matters, such as a pass threshold.

$$ICC = \frac{9}{9 + 4 + 3} \approx 0.56$$

Consistency

- › Raters may differ by a **constant offset**.
 - » Remove systematic rater offsets from the denominator.
- › **Use when:** relative differences between items matter.

$$ICC = \frac{9}{9 + 3} = 0.75$$

Hallgren (2012), [Computing Inter-Rater Reliability for Observational Data](#), "Common ICC variants." Illustrative variances; two-way random-effects, single-rater ICCs.



Disagreement tells you what to fix

Pattern in the disagreements	Repair to investigate
Raters split on " $x = 8/2$ "	Clarify the equivalent-answer anchor and add examples
Scores depend on seeing earlier tutor turns	Standardize the conversation context in the evidence packet
One rater's decisions change over time on repeated anchor cases	Check for drift; recalibrate against the frozen manual
Raters disagree about whether confirming an answer is acceptable help	Resolve the substantive definition or report separate dimensions



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- › Version the revised protocol, then assess it on **fresh, independently scored cases**.
- › Agreement reached by discussing these cases is part of development.



Build and report a reproducible evaluation

- › **1. Quantify uncertainty.**

- » Report the coefficient, variant, confidence interval, and interval method.

Hallgren (2012), [Computing Inter-Rater Reliability for Observational Data](#), study design, interpretation, and reporting.



Build and report a reproducible evaluation

- › **1. Quantify uncertainty.**

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- › **2. Report the scoring context.**

- » Raters, training, rubric version, evidence shown, and scoring procedure.

- » Sample size, case mix, category frequencies, rater overlap, and missingness.



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 - » Pilot → inspect disagreements → revise anchors and training → freeze → score fresh cases independently.



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- › **3. Improve the process in stages.**
 - » Pilot → inspect disagreements → revise anchors and training → freeze → score fresh cases independently.
- › **How much disagreement is tolerable?**
 - » Check how much system scores, failure rates, and system comparisons vary by rater.
 - » Aim to resolve meaningful improvements and agree closely on critical failures.
 - » Cutoffs are rough guides: consider case mix, uncertainty, and error consequences.



Validating an automated scorer



Three scorers judge the same conversations

Content: 180 saved student–tutor exchanges: 90 replies each from **Cohere North**, **Mini Code** and **Gemma 4 31B**.

Scorers: the answer-pattern rule, **Claude Sonnet 4.6** (strong), and **Ministral 14B** (weak). Both LLM judges run on Oppen.

Scorers compared	Agreement	Nominal α
Rule vs. Sonnet 4.6	91.7%	0.786
Rule vs. Ministral 14B	90.6%	0.758
Sonnet 4.6 vs. Ministral 14B	96.7%	0.917
All three	89.4%	0.821



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For all three, raw agreement means **unanimity**; α uses all within-conversation rater pairs.



Agreement leaves accuracy unresolved

α measures agreement between scorers. It cannot tell us **which decisions are correct**.

- › On **19 disputed conversations**, agreement cannot identify the correct verdict.
- › On **161 unanimous conversations**, all three could share an error.



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Establish a **held-out human reference** using the rubric and independent-rating protocol.

- › Include both tutors and cases where the automated scorers agree. Retain human disagreement alongside adjudicated labels.
- › Use that reference to estimate each scorer's **false-positive and false-negative rates**, then check whether the tutor comparison changes with the scorer.
- › Don't need to sample uniformly over examples
 - » e.g. p for unanimous cases; $2p$ for disputed (analyze with IPW)



Returning to the whole Theory of Change



A validated output score covers one link

An answer-leakage score measures whether the tutor **withholds the solution**.

The rest of the chain	Evidence still needed
Students use the tutor	Uptake and conversation logs
Students work through the problem	Their attempts and revisions
Students understand the material	An immediate, unaided assessment
Students retain and transfer understanding	A delayed, unaided assessment with new problems
Benefits come at an acceptable cost	Time, frustration, and differences by prior preparation



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Each measure needs its own validity argument. A causal design is also needed to attribute changes to the tutor.



Your project's measurement plan

Useful to plan specifics about what you will measure and how:

Specify	Example: independent understanding
Construct and definition	Explain, choose, and apply a method without assistance
Indicator and evidence	Choice, justification, and application on new problems
Unit and timing	One student's assessment, one week after tutor access ends
Scoring rule	Separate 0–2 ratings for each dimension, with anchors
Validation evidence to collect	Coverage of course objectives; pilot responses; independent double-scoring



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Mark the links you will **leave untested** and narrow your intended claim accordingly.



Assignment 2: ethics and responsible evaluation

Every team · Before session 6 (6 October) · 10% of the grade

Your design	Deliverable
Human-subjects experiment	Hertie self-assessment checklist : consent, risks, and data protection. Draft together, then submit through the official form.
Model-only evaluation	Evaluation data and responsibility statement , ≤ 2 pages: data and API terms, human labor, harmful outputs, and limits on claims.

Model-only teams using human raters or personal or scraped data also complete the relevant checklist sections.



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The **ethics steward coordinates**; teammates contribute and review. Start during the workshops while your design can change.



Next week's reflection memo

Bring a draft for **one central construct** in your project:

1. **A rubric:** definition, evidence shown to scorers, score levels, and scored examples on both sides of a difficult boundary.
2. **A validation plan:** who scores which cases, what stays held out, and which validity and agreement checks you will use. For an automated judge, specify the human reference and error checks.
3. **A measurement map:** where this score sits in your Theory of Change and what your design leaves unmeasured.

Use Jacobs & Wallach for the construct–indicator link, Hallgren for rater reliability, and Pangakis et al. for automated-scorer validation.

